Tutorial 11: Clustering

ENVX2001 – Applied Statistical Methods

Semester 1

Dissimilarity Matrices and Clustering

We are going to analyse the bird assemblage data from the lecture.

Setup

Load the vegan package:

CODE

```r
library(vegan)
```

Load the data

Read the data:

CODE

```r
macnally <- read.csv("data/macnally.csv", row.names = 1)
head(macnally, 1)
```

OUTPUT

```
           HABITAT V1GST V2EYR V3GF V4BTH V5GWH V6WTTR V7WEHE V8WNHE V9SFW
Reedy Lake   Mixed   3.4     0    0     0     0      0      0   11.9   0.4
           V10WBSW V11CR V12LK V13RWB V14AUR V15STTH V16LR V17WPHE V18YTH V19ER
Reedy Lake       0   1.1   3.8    9.7      0       0   4.8    27.3      0   5.1
           V20PCU V21ESP V22SCR V23RBFT V24BFCS V25WAG V26WWCH V27NHHE V28VS
Reedy Lake      0      0      0       0     0.6    1.9       0       0     0
           V29CST V30BTR V31AMAG V32SCC V33RWH V34WSW V35STP V36YFHE V37WHIP
Reedy Lake    1.7   12.5     8.6   12.5    0.6      0    4.8       0       0
           V38GAL V39FHE V40BRTH V41SPP V42SIL V43GCU V44MUSK V45MGLK V46BHHE
Reedy Lake    4.8   26.2       0      0      0      0    13.1     1.7     1.1
           V47RFC V48YTBC V49LYRE V50CHE V51OWH V52TRM V53MB V54STHR V55LHE
Reedy Lake      0       0       0      0      0     15     0       0      0
           V56FTC V57PINK V58OBO V59YR V60LFB V61SPW V62RBTR V63DWS V64BELL
Reedy Lake      0       0      0     0    2.9      0       0    0.4       0
           V65LWB V66CBW V67GGC V68PIL V69SKF V70RSL V71PDOV V72CRP V73JW
Reedy Lake      0      0      0      0    1.9    6.7       0      0     0
           V74BCHE V75RCR V76GBB V77RRP V78LLOR V79YTHE V80RF V81SHBC V82AZKF
Reedy Lake       0      0      0    4.8       0       0     0       0       0
           V83SFC V84YRTH V85ROSE V86BCOO V87LFC V88WG V89PCOO V90WTG V91NMIN
Reedy Lake      0       0       0       0      0     0     1.9      0     0.2
```

```
          V92NFB V93DB V94RBEE V95HBC V96DF V97PCL V98FLAME V99WWT V100WBWS
Reedy Lake      0     0       0      0     0    9.1        0      0        0
          V101LCOR V102KING
Reedy Lake        0        0
```

Calculate Bray-Curtis Dissimilarity

Calculate the Bray-Curtis dissimilarity:

```
CODE
Braydistance <- vegdist(macnally[, 3:102])
```

Hierarchical Clustering

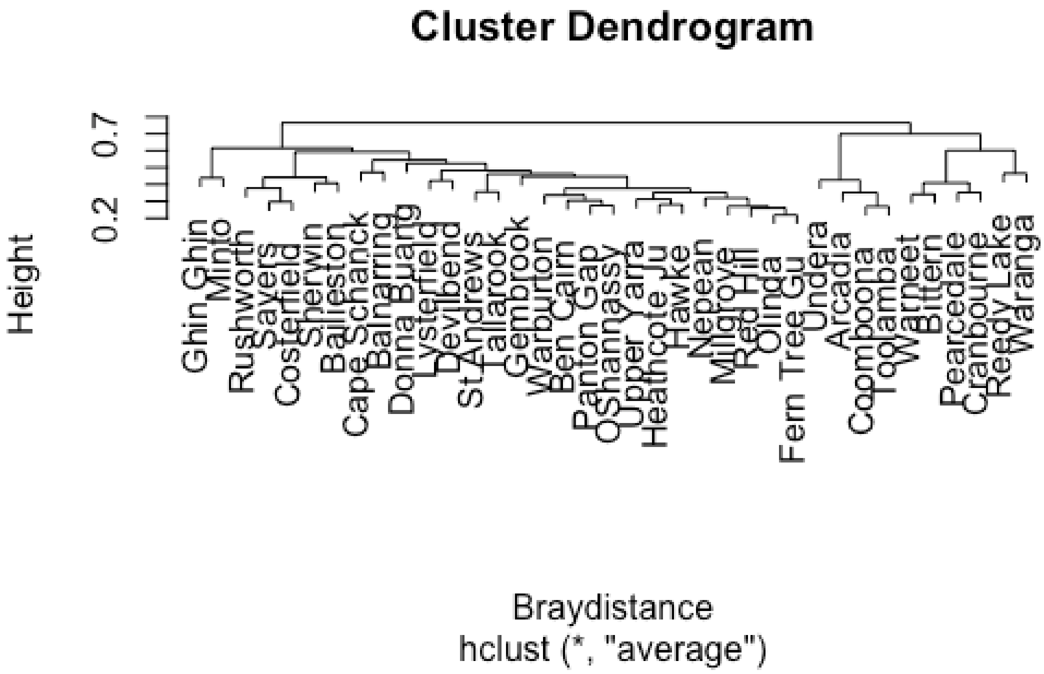
Apply hierarchical clustering using the UPGMA method ("average"):

```
CODE
hc <- hclust(Braydistance, method = "average")
```

Plot the Dendrogram

Plot the dendrogram:

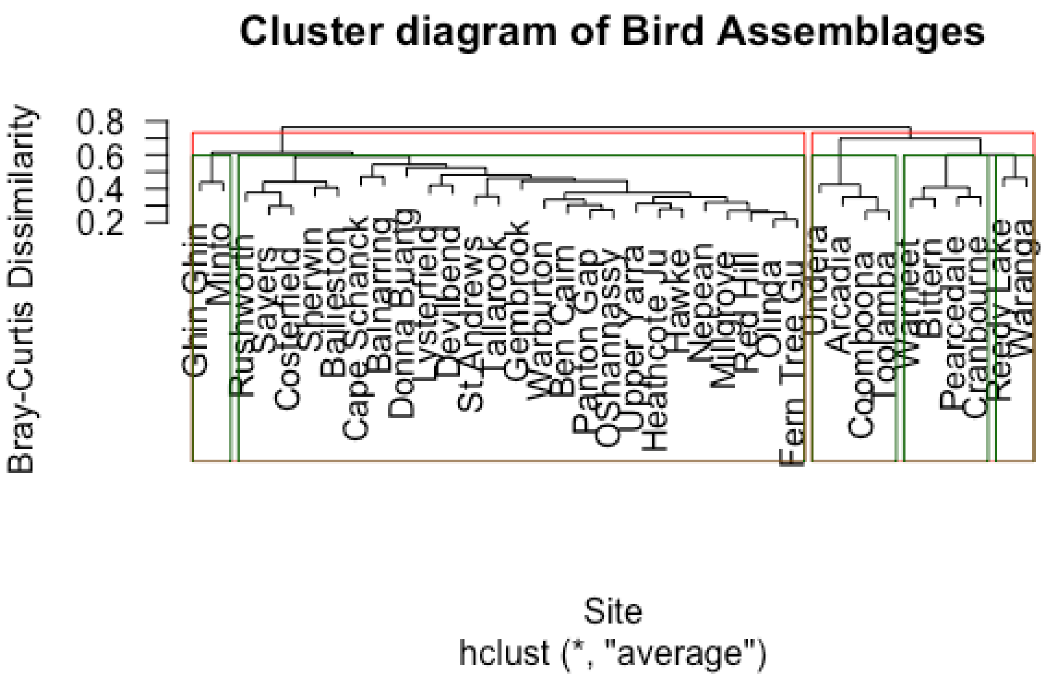
```
CODE
plot(hc)
```



Let's add some labels, and look at the potential number of clusters:

CODE

```
plot(hc, las = 1,
     main = "Cluster diagram of Bird Assemblages",
     xlab = "Site",
     ylab = "Bray-Curtis Dissimilarity")
rect.hclust(hc, 2, border = "red")
rect.hclust(hc, 5, border = "darkgreen")
```



The rectangles show potential groupings at different levels of the dendrogram:

- **Red rectangles**: 2 clusters
- **Green rectangles**: 5 clusters

The choice of how many clusters to use depends on your research question and the level of dissimilarity that is meaningful for your study.